

Amikosh Dube

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EDUCATION

Carnegie Mellon University

Master of Science in Computer and Civil Engineering

December 2026

Pittsburgh, PA

- Funded via Research Assistantship

- Cumulative GPA: 3.8/4.0

Purdue University

Bachelor of Science in Computer Engineering

May 2025

West Lafayette, IN

WORK EXPERIENCE

Autonomy Intern @ Aurora Flight Sciences

May 2026 – August 2026

- Designed and implemented a swappable rate-controller architecture (PID, RSLQR, OBLTR, AAOBLTR) in custom PX4 firmware, achieving 0.06–0.07 rad/s tracking error on roll/pitch under disturbance.
- Built a real-time, TensorRT-accelerated HLoc localization pipeline with a YOLOv11-n detector, removing the need for AprilTag markers, resulting in a 3x increase in inlier count and a 20% increase in pose-estimate accuracy.
- Formulated a closed-form analytical CBF safety filter for real-time obstacle avoidance in quadcopter guidance.
- Led a live quadcopter flight demonstration of the tuned AAOBLTR controller for Boeing and DARPA LINC program.

AirLabs Graduate Assistantship @ Carnegie Mellon University

August 2025 – Present

- Implemented multi-agent capability in AirStack, an open-source drone simulation platform built in Isaac Sim.
- Building an informative semantic path planning algorithm to navigate large construction sites.
- Implemented high-level object recognition and semantic mapping to contextualize 3D excavation surfaces for a Japanese construction firm, Shimizu, enabling volumetric differencing accurate to ± 5 cm.

Founder - Technical Lead @ Codex Labs LLC

May 2022 – August 2026

- Spearheaded the development of SWARMS, a cloud drone simulation platform bought by MIT Lincoln Laboratory.
- Led customer outreach, product demonstrations, and strategic development to expand the platform's adoption.

Machine Learning Software Engineering Intern @ SAAB Defense

May 2023 – August 2025 (Summers)

- Engineered a federated-learning AI pipeline that integrates multi-sensor data to produce situational intelligence.
- Trained a YOLO Drone Detector using 10,000 synthetic images, achieving 89% precision on real-world data.
- Developed a real-time, asynchronous object detection pipeline in Unreal Engine using C++, via NNE and OpenCV.
- Architected an automated image-annotation pipeline that accelerated YOLO training-data preparation.

Lead Drone SWARMS Researcher @ Purdue University

August 2021 – May 2025

- Led the research and development of SLAM algorithms for autonomous drone navigation in outdoor scenarios.
- Implemented semantic-based real-time obstacle avoidance using LiDAR and Cameras.
- Presented SLAM and swarm-control research 4 times at Purdue's Undergraduate Research Expo.

PROJECTS

CMU's Largest University-Built Drone: GoAERO Stage 1 and University Innovation Awardee **August 2025 – Present**

Graduate Projects: Top 10% in Deep Learning Kaggle competitions and MPC for Drones **August 2025 – Present**

VEX World Qualifier and Mentor: Four-time VEX Worlds Qualifier & top 50 ranked team **August 2017 – May 2025**

SKILLS

Programming Languages | C/C++, Python, SQL, Bash

AI & Robotics | Physical AI, Transformers, YOLO, SLAM, Path Planning, Sensor Fusion, PID/RSLQR/OBLTR/CBFs

Frameworks & Sim | Docker, ROS2, Gazebo, PX4, Unreal Engine, NVIDIA Isaac Sim, AirSim

Developer Tools | Git, GitHub, Scrum, JIRA, Confluence, Source Control, UML Diagrams